

Divergent effects of grass cover on soil infiltration patterns and water recharge in orchards: Taproot vs. fibrous root systems

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ABSTRACT

Grass cover is extensively adopted in orchards to control soil erosion, yet the mechanisms through which grass root architecture regulates soil infiltration and water redistribution remain poorly understood, particularly the dynamics between preferential flow (PF) and matrix flow (MF). This study elucidates how grass root architecture influences soil infiltration patterns and water recharge in orchard systems. Field experiments were conducted using a double-ring infiltrometer combined with a macropore-shielding method to evaluate infiltration behavior under three management practices: clean tillage (CT), fibrous-rooted grass cover (GCF), and tap-rooted grass cover (GCT). Results showed that grass cover significantly improved infiltration capacity and water recharge by 47.4 %–120.1 % and 38.2 %–107.2 %, respectively, with GCT exhibiting 1.46–1.72 times greater effectiveness than GCF. Furthermore, significant differences in infiltration patterns and water redistribution were observed between the grass types. The transition time from PF to MF dominance was shortest under CT (5 min), prolonged to 16 min under GCF, and PF remained dominant throughout the experiment under GCT. A dual regulatory mechanism of the two root architectures governing soil water flow paths was identified, leading to distinct water recharge patterns. In GCF, the enhancement of MF by soil structural improvement promoted topsoil water retention (0–20 cm). This layer accounted for 59.6 % of the total profile recharge, exceeding CT by a factor of 1.48. In contrast, GCT reinforced and sustained PF, increasing deeper water recharge (40–100 cm), which constituted 34.3 % of the total recharge and was 4.00 times that under CT. Therefore, selecting grass species based on root functions—taproot species for enhanced deep-water storage and drought resilience, and fibrous-root species for topsoil moisture conservation and erosion control—offers a strategy for targeted hydrological management in orchards. These findings advance the mechanistic understanding of root-mediated hydrological processes, and provide valuable insights for designing sustainable cover crop systems in rainfed agriculture.

1. Introduction

Orchard systems are a crucial component of global agriculture due to their high economic value. To meet the growing demand for fruits, the global orchard area increased by 22 % between 2000 and 2019 (FAO, 2022). Based on this trend, the global orchard area is projected to rise by 35 % by 2050 (Mason-D'Croz et al., 2019). Alarming, this expansion is increasingly challenged by the frequency of extreme climatic events under global warming. These conditions exacerbate the risks of soil

erosion and drought stress, thereby threatening the sustainable development of the fruit industry (Hou et al., 2024; Yuan et al., 2023). Soil infiltration following rainfall or irrigation is a crucial hydrological process, influencing surface runoff and serving as the primary source of soil water recharge (Cerdà, 1999; Wu et al., 2021; Liao et al., 2025a). Therefore, understanding the mechanisms of soil water infiltration and enhancing the infiltration capacity are essential for sustainable water resource management and climate resilience in orchard ecosystems.

In response to these challenges, grass cover has been widely adopted

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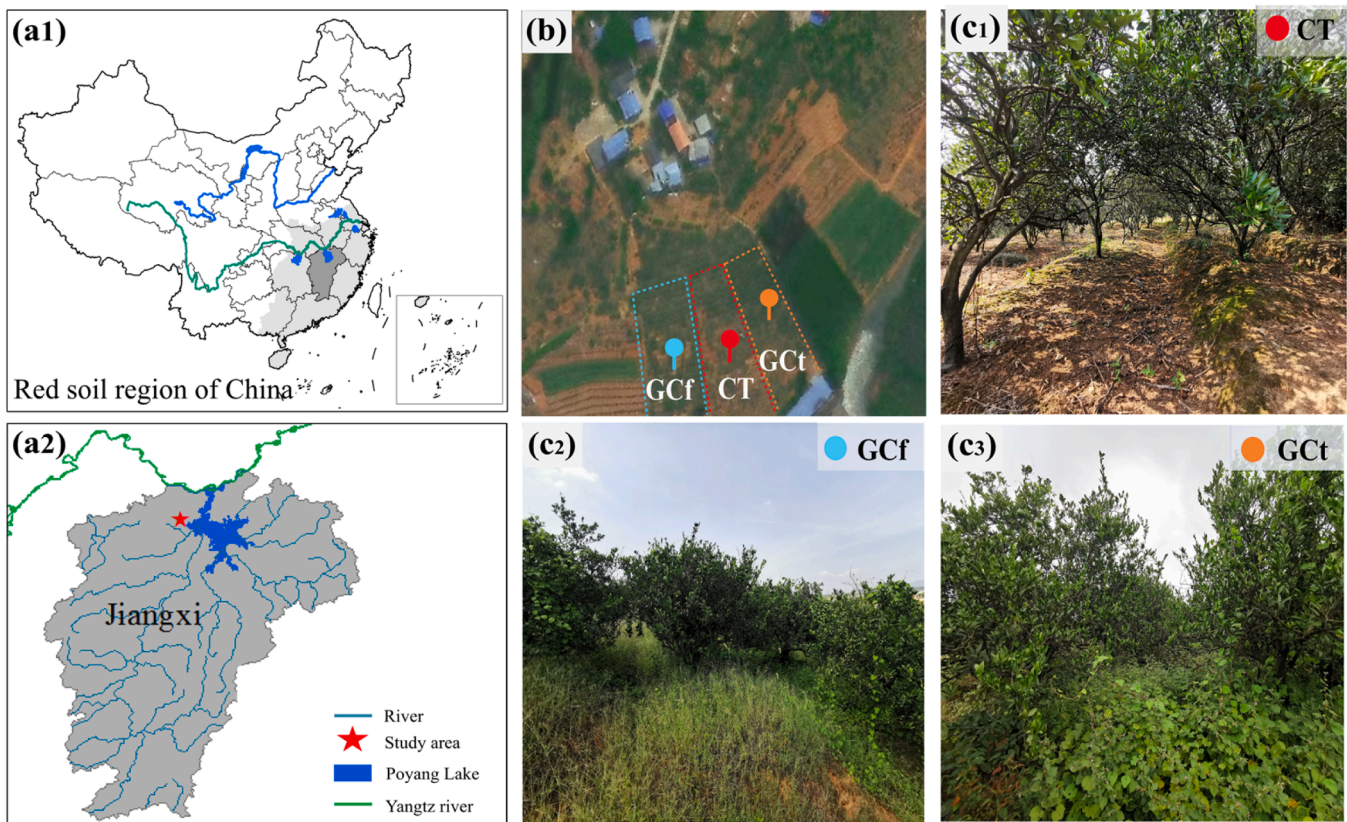


Fig. 1. Location of the study area (a), specific site location (b), and photographs (c) of the experimental orchards under different grass cover treatments. CT: clean tillage; GCf: fibrous-rooted grass cover; GCt: tap-rooted grass cover.

in orchards worldwide owing to its environmental benefits (Ren et al., 2023). This practice enhances surface coverage and root density within the soil profile, thereby improving soil properties (Blanco-Canqui and Ruis, 2020; Novara et al., 2021; Ding et al., 2024). As a key component of the soil system, root architecture (i.e., taproot and fibrous root systems) plays a significant role in regulating soil infiltration capacity (Liu et al., 2020; Ali et al., 2024; Fang et al., 2024). Numerous studies have demonstrated the benefits of grass roots on soil hydrological functions, particularly in reducing surface runoff and improving overall infiltration performance (Lu et al., 2020; Shi et al., 2021; Xiao et al., 2024). Nevertheless, the predominant focus on total infiltration has often obscured a critical mechanistic detail: the internal partitioning of water between two fundamentally distinct flow types, preferential flow (PF) and matrix flow (MF). PF is characterized by rapid and nonequilibrium water movement through macropores and biopores, which bypasses the soil matrix and promotes groundwater recharge (Beven and Germann, 2013; Guo et al., 2019; Zhang et al., 2020). In contrast, MF involves slower, capillary-driven water movement through the soil matrix, governing water storage and availability in the root zone (Zhang et al., 2017; Dhiman et al., 2018; Zhao et al., 2022). These pathways represent divergent hydrological destinies for soil water, with PF acting as a deep drainage and MF as a shallow storage (Cui et al., 2022; Lu et al., 2025). Failure to distinguish between them can lead to oversimplified hydrological models and ineffective soil-water management strategies (Sanders et al., 2012; Pollacco et al., 2024; Krüger and Külls, 2025).

A common approach for incorporating PF and MF into soil water transport simulations is to use a dual-porosity model (Gerke and Van Genuchten, 1993). This approach involves two parameterizations of the Richards equation, one for the PF domain and another for the MF domain. A significant limitation of such models is their simplistic representation of root effects; they typically either assume homogeneous root distributions or ignore root architecture altogether when defining

the PF domain and the MF domain (Shao et al., 2017; Mair et al., 2022). Consequently, there is a pressing need to quantify how contrasting root architectures modulate PF/MF partitioning and to trace how this partitioning controls the vertical distribution of soil water recharge. Addressing this gap is essential for developing new mathematical models that can accurately assess the hydrological impacts of specific root systems and guide targeted management (Mair et al., 2023).

Various methods exist to quantify PF and MF, including dye tracing, breakthrough curves analysis, and geophysical techniques (Allaire et al., 2009; Naveed et al., 2016; Pollacco et al., 2024). Among these, the paired double-ring infiltrometer method offers a practical field-based approach for separately quantifying PF and MF infiltration rates (Zhang et al., 2019; Zhao et al., 2023; Liao et al., 2025b; Wen et al., 2025). In this method, one infiltrometer is covered with a layer of fine sand or soil-sand mixture to shield PF pathways, thereby enabling measurement of the MF infiltration rate. The PF infiltration rate and its contribution are then calculated from the difference between the total and MF infiltration rates. This approach is cost-effective, operationally simple, and provides reliable discrimination between PF and MF dynamics. Although validated in forest soils of arid and semi-arid regions, its application to orchard systems under different grass cover types—and specifically its use to decipher the role of root architecture in flow partitioning—remains limited. This knowledge gap hinders the design of precise cover crop strategies for specific ecohydrological objectives, such as optimizing drought resilience or erosion control.

To address these gaps, field paired-infiltration experiments were carried out in orchards with contrasting grass cover types (taproot vs. fibrous-root species). The specific objectives of this study were to (1) evaluate the effects of root architecture on PF and MF infiltration dynamics; (2) determine how these infiltration patterns govern the depth distribution of soil water recharge; and (3) elucidate the mechanisms by which root architecture regulates flow partitioning and water

Table 1

Basic site information for the experimental orchards under different grass cover treatments. Values are presented as mean $\pm$ standard deviation ($n = 5$).

Measures	Groundcover management	Grass species	Root system	Slope (°)	Age (year)	Canopy coverage (%)	Plant height (m)	Ground diameter (cm)	Initial soil moisture (%)
CT	Clean tillage	—	—	10–12	20	80.5 $\pm$ 4.2	3.2 $\pm$ 0.2	13.9 $\pm$ 2.5	22.15 $\pm$ 4.79
Gcf	Grass cover	<i>Digitaria sanguinalis</i> (L.) Scop.	Fibrous roots	10–12	20	76.7 $\pm$ 5.3	3.1 $\pm$ 0.3	14.1 $\pm$ 5.3	22.89 $\pm$ 5.17
GCt	Grass cover	<i>Urena lobata</i> L.	Taproots	10–12	20	78.8 $\pm$ 4.8	2.9 $\pm$ 0.2	14.7 $\pm$ 2.1	21.97 $\pm$ 6.69

redistribution. This study hypothesized that: (1) compared to fibrous-root species, taproot species would promote a stronger and more persistent PF-dominated regime due to their coarser and deeper root systems; and (2) a PF-dominated regime would direct a greater proportion of infiltrated water to deep soil layers, thereby enhancing deep soil water storage. The study aims to advance the mechanistic understanding of root-mediated hydrological processes and support the design of targeted soil and water conservation practices for orchards under changing climatic conditions.

2. Materials and methods

2.1. Study area

The red soil region of China is a major area for citrus production, and more locations are being exploited and planted with citrus crops. Over the past six decades, this region has experienced frequent floods and droughts due to climate change and human activities (Bi et al., 2023; Zhong et al., 2025). In 2022, the region experienced its most extreme drought event since 1961, resulting in a water storage loss of approximately 158 $\pm$ 22 km³ (Duan et al., 2024) and posing a serious threat to agricultural productivity. Consequently, the Yangou watershed (29°16' N to 29°17' N, 115°42' E to 115°43' E), located in De'an County of Jiangxi Province, was selected as the study area in the central part of the red soil region (Fig. 1a). The study area is characterized by a typical subtropical humid monsoon climate with distinct wet and dry seasons. The mean annual precipitation is 1469 mm, and approximately 70 % or more of the total precipitation occurs from March to August. The mean annual temperature is approximately 16.7 °C. The region is characterized by shallow hilly land, with slopes ranging from 5° to 25°. The dominant soil type in the region is red clay soil, which is formed by the decomposition of Quaternary sediments, and it reaches over 100 cm in depth. Red clay soil is classified as an Ultisol in the USDA soil taxonomy system.

The population density, topography and land-use changes in the study area are representative of the red soil region of China. Hilly mountains are the most widely distributed landform in this region, and slope agriculture is commonly performed. With population growth and economic development, many sloping lands between 8° and 15° have been transformed into citrus orchards due to their high economic benefits for local farmers (Duan et al., 2020). Consequently, citrus orchards represent the main land use type in the study area. However, severe soil erosion often occurs in citrus orchards due to intense rainfall and clean tillage management practices (Niu et al., 2021). To mitigate water erosion, grass cover management, such as understory grass planting and maintenance, is widely implemented (Duan et al., 2020; Ren et al., 2023). Therefore, the study area is typical of the red soil region in terms of climate, soil, topography, and land use types.

2.2. Experimental design

In the study area, citrus trees (*Citrus reticulata* Blanco) were planted in the 2000s with a spacing of 3 m $\times$ 3 m. Following plantation, three ground cover management treatments were implemented: clean tillage (CT), fibrous-rooted grass cover (Gcf), and tap-rooted grass cover (GCt). These treatments were implemented in three adjacent citrus orchards,

each covering an area of approximately 2000 m² (Fig. 1b). The maximum distance between any two experimental orchards did not exceed 1 km, ensuring similar site conditions across them. More detailed information about the three experimental orchards is presented in Table 1.

In the CT treatment, glyphosate herbicide was applied every two months to eliminate weeds. In the two grass-covered treatments (Gcf and GCt), understory herbaceous plants were allowed to grow naturally and were mowed annually. The predominant grass species with fibrous roots and taproots were crabgrass (*Digitaria sanguinalis* (L.) Scop.) and caesarweed (*Urena lobata* L.), respectively. Both are perennial herbaceous plants widely distributed in orchards across the red soil region of China, recognized for their soil-water conservation benefits and medicinal value. Apart from the specified ground cover treatments, all other agronomic practices, including pruning, pest and disease control, frost protection, and fruit harvesting, were consistently maintained across all treatments (Fig. 1c).

Within each experimental orchard, a 20 m $\times$ 20 m subplot was established. Along the diagonal of each subplot, five 3 m $\times$ 3 m sampling quadrats were randomly positioned, with each quadrat's corner anchored by a citrus tree. Two double-ring infiltrometers were deployed within inter-tree spaces of each quadrat (Fig. S1a): one for measuring total infiltration and another for assessing matrix infiltration. To avoid hydrological interference, the two infiltration points were spaced 0.7 m to 0.8 m apart (Zhang et al., 2019; Zhao et al., 2023; Wen et al., 2025; Liao et al., 2025b). Within each experiment orchard, five total infiltration measurements and five matrix infiltration measurements were conducted, resulting in a total of 30 infiltration measurements (3 treatments $\times$ 2 infiltration types $\times$ 5 replications).

2.3. Soil total infiltration and matrix infiltration measurement

All infiltration experiments were conducted from July to August 2023. Prior to the infiltration tests, the experimental sites had received no precipitation for at least two weeks. Both total infiltration and matrix infiltration were measured using a double-ring infiltrometer with inner and outer ring diameters of 30 cm and 60 cm, respectively (Fig. S1b). After selecting a level location, the aboveground part of herbs and the litter layer were carefully removed to expose the soil surface. Then, the double rings were gently and vertically inserted 2–3 cm into the soil to minimize soil disturbance (Zhang et al., 2017; Zhao et al., 2022). Loose soil materials were placed outside of both the inner and outer rings to seal water from leaking outside of the rings. The soil surface inside both rings was covered with a 5 cm $\times$ 5 cm cardboard to protect it from being directly scoured during the water addition process. A Brilliant Blue FCF dye solution (4 g L⁻¹) and clean water were rapidly and simultaneously added to the inner and outer rings, respectively, until a water depth of 5 cm was reached at the same time in both rings. The falling-head method was used to measure water infiltration process (Wu et al., 2021; Haruna et al., 2022). The time required for the water level in the inner ring to drop by 5 mm was recorded with a stopwatch. This process continued until the infiltration rate stabilized for three consecutive readings, which was considered the steady infiltration rate (Guo et al., 2019; Liu et al., 2020; Liao et al., 2025a). When the water level dropped to 1 cm depth, the inner and outer rings were refilled to 5 cm with the dye solution and clean water, respectively, which means that 4 cm of

dye solution and water were added each time. The water levels in the inner and outer rings were maintained consistently throughout the infiltration process.

For the matrix infiltration measurement, the soil surface was similarly cleared of herbs and the litter layer before setting up the double rings. Previous studies have indicated that a soil-sand mixture as a covering material can effectively block the connectivity of macropores (PF paths) on the soil surface without affecting the matrix infiltration process (Zhang et al., 2017; Zhao et al., 2022). Initially, the infiltration rate is controlled by the soil-sand mixture layer with higher conductivity. However, when the wetting front reaches and penetrates the underlying soil, the infiltration rate decreases until it approximately equals that of the soil matrix. At this point, the soil matrix with lower conductivity controls the infiltration process (Allaire et al., 2002; Zhang et al., 2017). Therefore, a thin layer of soil-sand mixture was placed on the soil surface to block macropores and ensure water infiltrating only through the soil matrix. Based on preliminary experiments, a 3 cm thick layer of a soil-quartz sand (80 mesh) mixture (mass ratio 2:3) was uniformly spread over the surface inside the inner and outer rings (Fig. S1b) (Zhao et al., 2022). The subsequent procedure for the matrix infiltration measurement was identical to that for the total infiltration.

2.4. Calculation of preferential flow process

The total infiltration measured from the inner ring of the double-ring infiltrometer is partitioned into PF and MF. Thus, the PF infiltration rate can be obtained from the difference between the total infiltration rate and the MF infiltration rate (Zhang et al., 2019; Zhao et al., 2023; Wen et al., 2025). Specifically, total infiltration at point 1 was measured directly using the double ring infiltrometer without surface treatment. MF infiltration at point 2 was measured by spreading a soil-sand mixture layer on the soil surface inside the inner and outer rings. Due to the uniformity of MF at a small scale (Das Gupta et al., 2006; Zhang et al., 2017), the MF infiltration rate at point 1 can be replaced by the value measured at the adjacent point 2. Therefore, the PF infiltration rate at point 1 was calculated by the following formula:

$$i_{1p} = \frac{dQ_{1p}}{dt} = \frac{dQ_{1t}}{dt} - \frac{dQ_{1m}}{dt} = \frac{dQ_{1t}}{dt} - \frac{dQ_{2m}}{dt} = i_{1t} - i_{2m} \quad (1)$$

where i_{1p} is the PF infiltration rate at point 1 (mm h^{-1}), i_{1t} is the total infiltration rate at point 1 (mm h^{-1}), and i_{2m} is the MF infiltration rate at point 2 (mm h^{-1}); Q_{1t} represents the total infiltration amount at point 1 (mm), Q_{1p} represents the PF infiltration amount at point 1 (mm), Q_{1m} and Q_{2m} represent the MF infiltration amount at point 1 and point 2, respectively (mm); and t represents the infiltration time (h).

Water temperature greatly influences the soil infiltration rate (Suwardji and Eberbach, 1998). To ensure the comparability of the results, the measured infiltration rates at different water temperatures were converted to standard infiltration rates under 10 °C using the following equation:

$$i_{10} = \frac{i}{(0.7 + 0.03T)} \quad (2)$$

where i_{10} is the infiltration rate at water temperature of 10 °C (mm h^{-1}), i is the infiltration rate at any water temperature (mm h^{-1}), and T is the water temperature during each experiment (°C).

2.5. Soil profile excavation and soil samples

After the infiltration experiments, plastic film was placed over the wetted areas to minimize surface evaporation. Three parallel vertical soil profiles (60 cm wide $\times$ 100 cm deep) were excavated about 24 h after finishing each infiltration experiment (Guo et al., 2019; Liu et al., 2020). The profiles were excavated tangent to the inner ring and at positions corresponding to 1/4 and 1/2 of the inner ring's diameter.

After completing the vertical profile at the 1/2 diameter position, horizontal profiles were excavated at depths of 5, 10, 20, 40, and 60 cm within the remaining soil body (Fig. S1c). After each excavation, a brush and a hand-held electric fan were used to clean the fallen soil particles, thereby exposing the stained and wetted areas. After leveling the soil profile, the registration ruler was placed horizontally and vertically along the profile. The wetted soil profiles were photographed with a digital camera. The photographs were then transformed into grayscale images and binarized to obtain black-and-white biphasic pictures using Adobe Photoshop CC 2017. The binary images were digitized by using the "Bitmap Analysis" function in Image-Pro Plus 6.0. Finally, the Image-Pro Plus 6.0 software was used to reduce the image noise and obtain binary images, in which the stained area was white and the unstained area was black. The final image was transformed into a binary matrix data to analyze the preferential flow characteristics (Dai et al., 2021; Du et al., 2025).

From each horizontal profile, three replicated soil samples were collected from the same soil layer and combined into one composite sample. These soil samples were used to determine water-stable aggregate and organic matter content in the laboratory. Two cutting rings (200 cm^3 in volume) were used to collect undisturbed soil samples from each layer with three replications for subsequent analysis of soil bulk density, porosity, and root morphological parameters. Moreover, after excavating the vertical soil profile tangent to the ring, three replicated soil samples were collected from another soil profile at depths of 0–5, 5–10, 10–20, 20–40, 40–60, and 60–100 cm, for measurement of initial soil moisture. Roots were separated using a 0.25 mm sieve, rinsed with water, and dried with a nonwoven fabric. Then, they were scanned on a flatbed scanner at 300 dpi using an Epson V900 photo scanner from Japan. Root morphological traits including root length density (RLD), root surface area density (RSD), and root volume density (RVD) were quantified using WinRHIZO software (Liu et al., 2020; Wang et al., 2023). Finally, the roots were oven-dried at 60 °C to a constant weight to determine root mass density (RMD).

Soil water content was measured gravimetrically by drying soil samples at 105 °C to constant weight. For each infiltration event, soil water recharge (SWR) in the vertical profile was calculated using the following formula (Wu et al., 2021; Liao et al., 2025a):

$$\text{SWR} = (\text{SWC}_1 - \text{SWC}_0) \times h \times \text{BD} \quad (3)$$

where SWC_1 is the soil water content after infiltration for a specific soil layer; SWC_0 is the initial soil water content before infiltration for a specific soil layer; h is the depth of a specific soil layer (mm); and BD is the bulk density of a specific soil layer (g cm^{-3}).

2.6. Data analysis

One-way analysis of variance (ANOVA) with a least significant difference (LSD) test was performed to evaluate differences in soil infiltration properties, soil water recharge (SWR), and root traits among the experimental treatments. Two-way ANOVA with an LSD test was employed to assess the impacts of experimental treatments, soil depth and their interactions on soil physicochemical properties. Prior to ANOVA, Shapiro's and Levene's tests were used to check the data normality and homogeneity, respectively. If the assumptions were violated, the Kruskal-Wallis ANOVA test with a Dunn's test was utilized. Pearson correlation analysis was performed using the 'corrplot' package to examine the relationships between the response variables (PF and MF infiltration rates) and explanatory variables (root traits and soil physicochemical properties). Variance partitioning analysis (VPA) was performed using the 'vegan' package to quantify the interactive and relative contributions of three categories of explanatory variables (root traits, soil structure, and organic matter) to the variation in the response variables. Partial least squares path modeling (PLS-PM) was used to analyze the direct and indirect effects of root characteristics on SWR

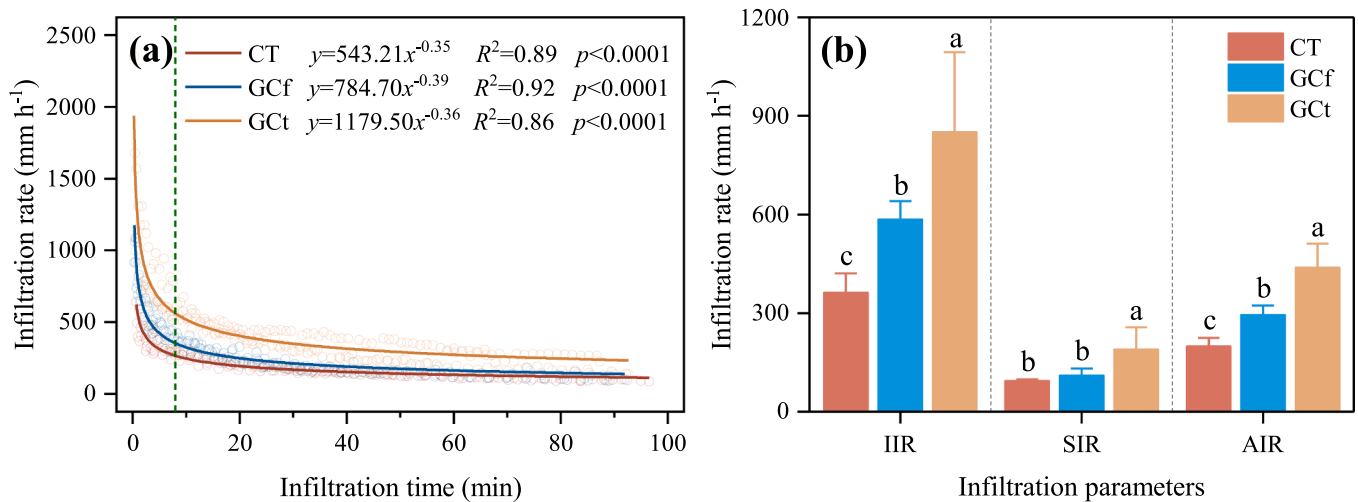


Fig. 2. Temporal variation of the total infiltration rate (a) and comparison of infiltration parameters (b) under different grass cover treatments. IIR: initial infiltration rate; SIR: steady-state infiltration rate; AIR: average infiltration rate. CT: clean tillage; GCf: fibrous-rooted grass cover; GCt: tap-rooted grass cover. Bars with different lowercase letters indicate significant differences between treatments ($p < 0.05$).

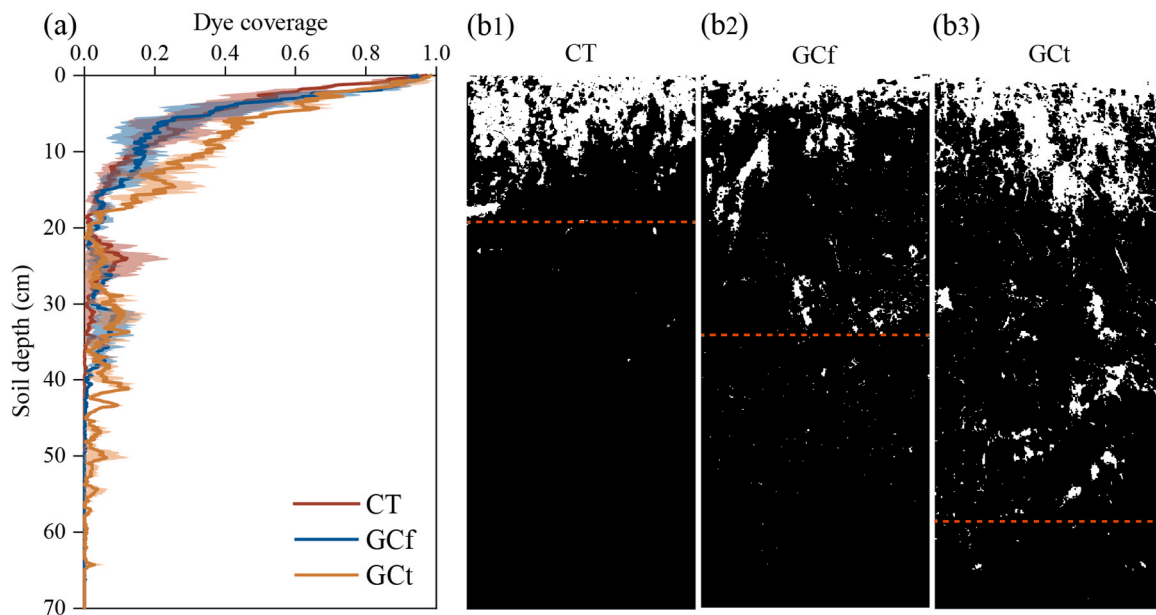


Fig. 3. Dye staining coverage (a) and corresponding binary images (b) of soil vertical profiles following total infiltration tests under different grass cover treatments. The solid lines and the shaded ribbons represent the average dye coverage and standard deviation, respectively. CT: clean tillage; GCf: fibrous-rooted grass cover; GCt: tap-rooted grass cover.

across different soil layers. PLS-PM was chosen because it relies on a series of ordinary least squares regressions and has minimal sample size requirements. Furthermore, PLS-PM can achieve high statistical power when other methods fail to converge or provide inadmissible solutions. PLS-PM models were evaluated based on the overall goodness-of-fit index and the R^2 value for SWR. Statistical significance was determined at $p < 0.05$, $p < 0.01$, and $p < 0.001$. All statistical analyses and plotting were performed in R version 4.5.1, and the path analysis was conducted using the 'plspm' package.

3. Results

3.1. Soil infiltration capacity and patterns

By the end of the measurement period, the infiltration rates for all treatments showed no significant change and were approaching a steady

state. The total infiltration rate decreased rapidly during the initial stage (0–8 min) across all treatments, followed by a more gradual decline until stabilization (Fig. 2a). Relative to clean tillage (CT), both grass cover treatments (GCf and GCt) significantly enhanced soil infiltration capacity ($p < 0.05$), increasing initial (IIR), steady (SIR), and average (AIR) infiltration rates by 61.4 %–134.9 %, 17.4 %–102.2 %, and 47.4 %–120.1 %, respectively (Fig. 2b). Notably, GCt was significantly more effective than GCf in improving soil infiltration capacity ($p < 0.05$), with IIR, SIR, and AIR values 1.46, 1.72, and 1.49 times higher, respectively (Fig. 2b).

Soil profiles excavated following total infiltration tests showed distinct dye staining, whereas no staining was observed in profiles after the matrix flow (MF) tests (Fig. S2). This confirmed that preferential flow (PF) paths were effectively blocked during MF measurements, thereby ensuring the assessment of true matrix flow infiltration. The PF infiltration depth varied substantially among treatments, being greatest

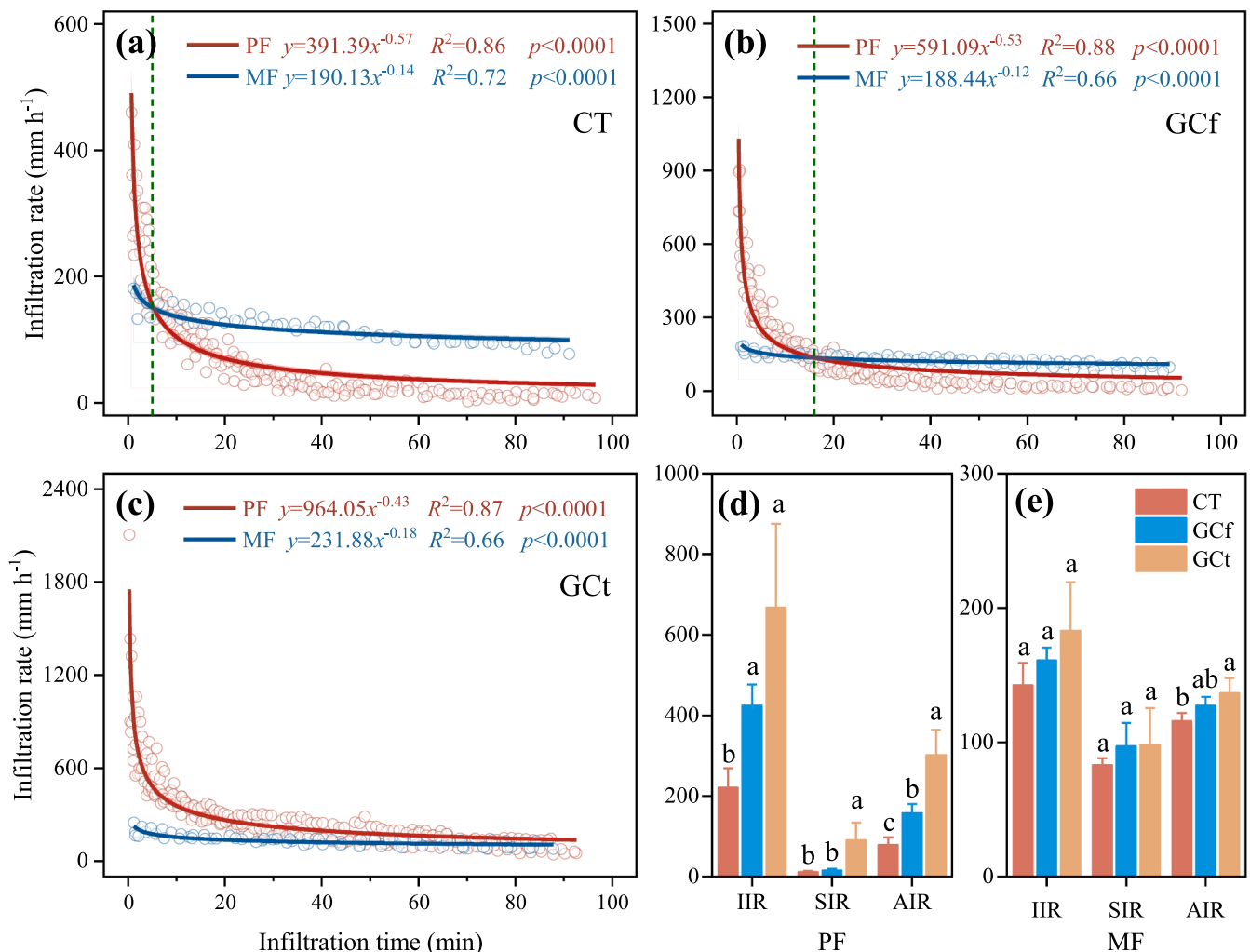


Fig. 4. Temporal variation of preferential flow (PF) and matrix flow (MF) infiltration rates under different grass cover treatments. The vertical dashed lines indicate the time at which the PF and MF infiltration rates became equal. CT: clean tillage; GCf: fibrous-rooted grass cover; GCt: tap-rooted grass cover. Bars with different lowercase letters indicate significant differences between treatments ($p < 0.05$).

under GCt (57.6 cm), intermediate under GCf (38.5 cm), and least under CT (18.4 cm) (Fig. 3). Notably, the dominant flow regime differed significantly among treatments, reflecting the profound impact of grass cover on water transmission pathways. In CT and GCf, flow transitions from initial PF dominance to sustained MF dominance within minutes (5 and 16 min, respectively). After this transition point, the relative contribution of PF to the ongoing infiltration flux becomes minor compared to MF. PF infiltration rates consistently exceeded MF infiltration rates under GCt throughout the experiment. However, the absolute difference between PF and MF diminishes over time (Fig. 4). Specifically, during the initial stage, PF infiltration rates were 1.55–3.59 times greater than MF infiltration rates, accounting for 60.4 %–78.1 % of total infiltration. In contrast, during the steady state, MF infiltration rates were 1.22–7.48 times higher than PF infiltration rates, constituting 53.5 %–87.8 % of total infiltration (Table 2).

PF infiltration rates differed significantly between grass cover types ($p < 0.05$; Fig. 4d). Compared to CT, grass cover markedly increased PF infiltration rates, with IIR, SIR, and AIR increasing by 92.1 %–202.3 %, 33.8 %–678.0 %, and 101.3 %–283.9 %, respectively. GCt demonstrated substantially greater enhancement of PF than GCf, exhibiting IIR, SIR, and AIR values that were 1.57, 5.82, and 1.91 times higher. In contrast, although MF infiltration rates under grass cover were numerically higher than under CT, the differences were not statistically significant ($p > 0.05$). In terms of cumulative infiltration amount, PF was

the dominant process under GCt, representing 63.5 % of the total infiltration. Conversely, MF dominated under CT and GCf, accounting for 70.4 % and 60.9 %, respectively (Table 2).

3.2. Soil water recharge

Under identical infiltration conditions, significant differences in soil water recharge (SWR) were observed among treatments ($p < 0.05$; Fig. 5a). Both grass cover types showed significantly greater total SWR throughout the soil profile compared to CT, with increases of 38.2 % for GCf and 107.2 % for GCt. While SWR differences between grass cover types were minimal in the topsoil (0–20 cm), they became more pronounced with increasing soil depth (Fig. 5b). Compared to CT, GCf and GCt enhanced deep-soil (40–100 cm) SWR by factors of 1.17 and 4.00, respectively. The topsoil SWR under GCf was 1.48 times that under CT, and the water recharge under GCt reached the deepest soil layer (60–100 cm). Furthermore, the proportion of SWR (PSWR) distribution also varied with soil depth (Fig. 5b). Under GCt, water recharge was enhanced in the deep soil, with a PSWR of 34.3 %. This value was 1.97 and 2.14 times greater than that under CT and GCf, respectively. Conversely, the PSWR in the topsoil under GCt (38.3 %) was lower than under CT (55.2 %) and GCf (59.6 %).

Table 2

Characteristics of preferential flow (PF) and matrix flow (MF) infiltration under different grass cover treatments.

Infiltration parameters	Treatment	PF/MF	PF/TI	MF/TI
IIR (mm h ⁻¹)	CT	1.55 ± 0.28c	0.604 ± 0.042b	0.396 ± 0.042a
	GCf	2.64 ± 0.33b	0.724 ± 0.024a	0.276 ± 0.024b
	GCT	3.59 ± 0.39a	0.781 ± 0.018a	0.219 ± 0.018b
	CT	0.14 ± 0.04b	0.122 ± 0.028b	0.878 ± 0.028a
SIR (mm h ⁻¹)	CT	0.16 ± 0.03b	0.138 ± 0.020b	0.862 ± 0.020a
	GCf	0.91 ± 0.31a	0.465 ± 0.096a	0.535 ± 0.096b
	GCT	0.67 ± 0.14c	0.400 ± 0.054c	0.600 ± 0.054a
	CT	1.24 ± 0.18b	0.552 ± 0.037b	0.448 ± 0.037b
AIR (mm h ⁻¹)	CT	2.19 ± 0.32a	0.685 ± 0.030a	0.315 ± 0.030c
	GCf	0.43 ± 0.10c	0.296 ± 0.049c	0.704 ± 0.049a
	GCT	0.65 ± 0.12b	0.391 ± 0.047b	0.609 ± 0.047b
	CT	1.75 ± 0.18a	0.635 ± 0.025a	0.365 ± 0.025c

Note: Values are means ± standard deviation. PF: preferential flow; MF: matrix flow; TI: total infiltration. PF/MF: the ratio of infiltration rate between PF and MF; PF/TI: the ratio of infiltration rate between PF and TI; MF/TI: the ratio of infiltration rate between MF and TI. IIR: initial infiltration rate; SIR: steady-state infiltration rate; AIR: average infiltration rate; CIA: cumulative infiltration amount. CT: clean tillage; GCf: fibrous-rooted grass cover; GCT: tap-rooted grass cover. Different lowercase letters indicate significant differences among the mean values of different grass cover treatments ($p < 0.05$).

3.3. Root characteristics and soil physicochemical properties

Root characteristics differed significantly between grass cover types ($p < 0.05$; Fig. 6). Both GCf and GCT exhibited significantly greater root length density (RLD), root surface area density (RSD), root volume density (RVD), and root mass density (RMD) than CT across most soil depths (except for RVD at 40–100 cm). This indicated that grass cover

effectively promoted root growth. Root spatial distribution patterns differed considerably between grass cover types. GCf showed the highest root density in the topsoil (0–20 cm), which decreased rapidly with depth, whereas GCT maintained high root densities in both the topsoil and subsoil (20–40 cm), with significantly higher RMD in the subsoil than in GCf (Fig. 6d; $p < 0.05$). Although GCT exhibited higher root density in the deep soil (40–100 cm) than GCf, the difference was not significant ($p > 0.05$).

Both grass cover types reduced soil bulk density while increasing total porosity and soil organic matter content compared to CT (Fig. 7a–c). GCT enhanced soil porosity in both the topsoil and subsoil, while GCf only increased porosity in the topsoil. Soil organic matter was significantly higher under GCf than under CT in the topsoil, but higher under GCT in both the subsoil and deep soil. Similarly, both grass cover types significantly promoted macroaggregate formation and improved aggregate water stability (Fig. 7d–e). The positive effect on macroaggregates was restricted to the topsoil for GCf but extended into the subsoil and even the deep soil for GCT. Grass cover treatments also exhibited higher initial soil water content in the deep soil compared to CT (Fig. 7f). Regarding soil texture, grass cover led to a lower silt content than CT, while clay and sand contents were not significantly affected (Fig. 7g–i; $p > 0.05$).

3.4. Key factors influencing infiltration patterns and soil water recharge

Pearson correlation analysis indicated PF infiltration rates were significantly correlated with root traits, soil organic matter (SOM), bulk density (BD), total porosity (TP), and sand content ($p < 0.01$), but not with macroaggregates (Macro) and mean weight diameter (MWD) ($p > 0.05$). In contrast, MF infiltration rates were strongly correlated with SOM and BD ($p < 0.01$), and significantly correlated with Macro, MWD, and sand content ($p < 0.05$), but showed no significant correlation with root traits ($p > 0.05$) (Fig. 8a). The VPA results revealed that root traits explained a substantially larger proportion of the variance in PF (31.5 %) than in MF (12.5 %) (Fig. 8b–c). In contrast, SOM alone explained a slightly larger fraction of the variance in MF (11.0 %) than in PF (9.2 %). These results demonstrate that PF and MF are controlled by distinct dominant factors.

Path analysis showed acceptable model fits, with goodness-of-fit indices ranging from 0.56 to 0.60 (Fig. 9). The path models exhibited greater explanatory power for SWR in the deep soil ($R^2=0.73$) than in

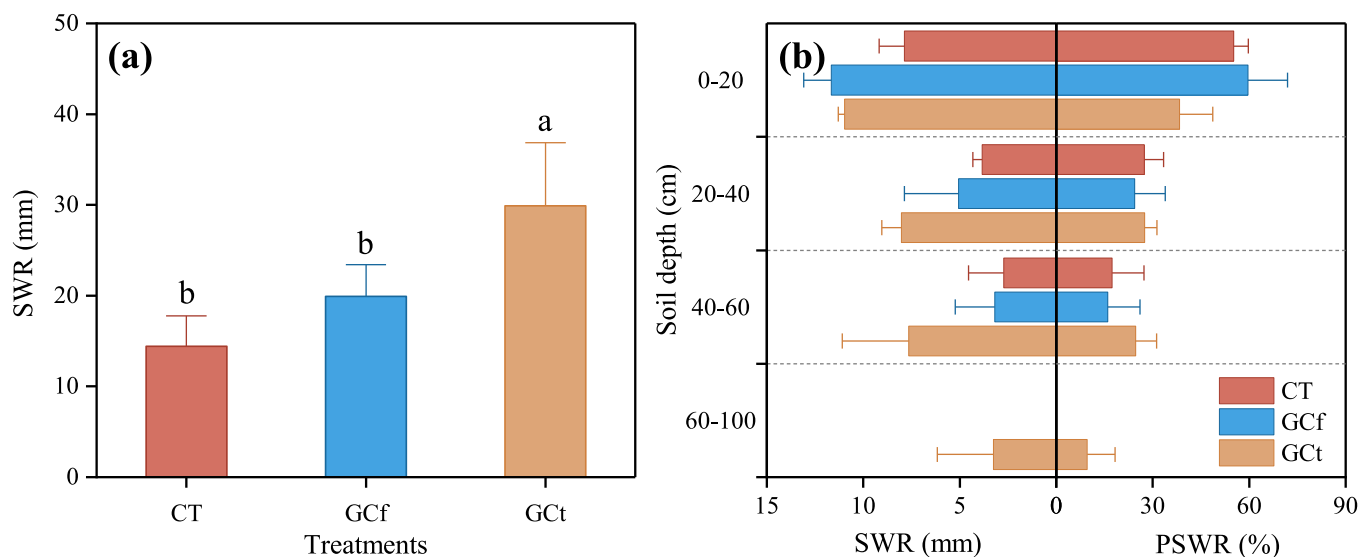


Fig. 5. Total soil water recharge (SWR) over the entire profile (a) and the proportion of the soil water recharge (PSWR) within each soil layer (b) under different grass cover treatments. CT: clean tillage; GCf: fibrous-rooted grass cover; GCT: tap-rooted grass cover. Bars with different lowercase letters indicate significant differences between treatments ($p < 0.05$).

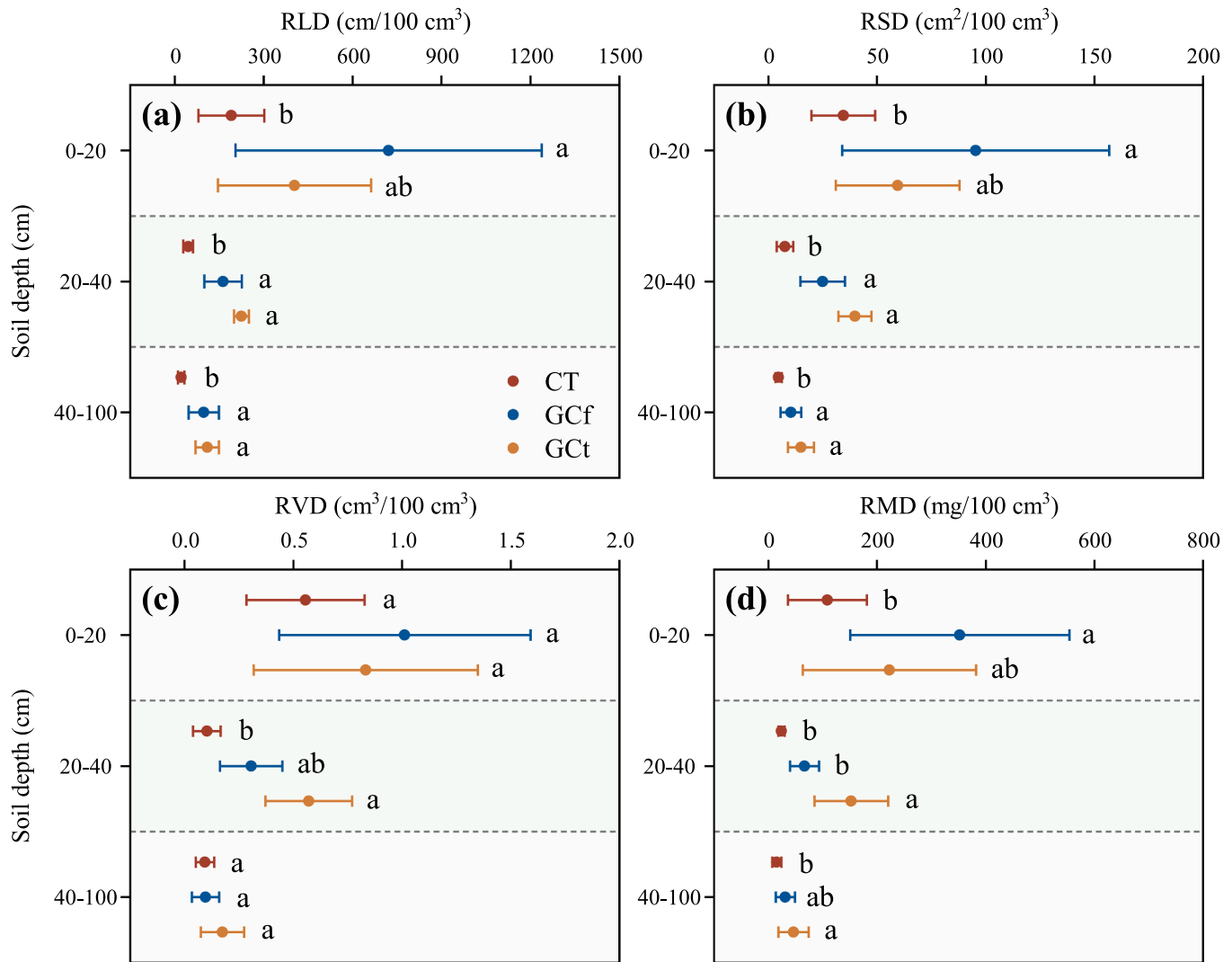


Fig. 6. Root characteristics at different soil depths under different grass cover treatments. Data are presented as means $\pm$ standard deviation (SD). RLD: root length density; RSD: root surface area density; RVD: root volume density; RMD: root mass density. CT: clean tillage; GCf: fibrous-rooted grass cover; GCt: tap-rooted grass cover. Different lowercase letters indicate significant differences between treatments at the same soil depth ($p < 0.05$).

the subsoil ($R^2=0.63$) and topsoil ($R^2=0.45$). The mechanism by which root traits influenced soil infiltration and SWR varied with soil depth. In the topsoil, roots primarily improved soil structure, thereby increasing SWR by enhancing MF (path coefficient = 0.67, $p < 0.05$; Fig. 9a). In the subsoil and deep soil, PF had the strongest direct effect on SWR (path coefficients = 0.59 and 0.87, respectively; $p < 0.05$), and root traits were the primary driver of PF at these depths (path coefficients = 0.74 and 0.82, respectively; $p < 0.05$) (Fig. 9b–c). Moreover, cross-loading effects confirmed that the direct effect of PF on SWR was strongest in the deep soil, whereas the direct effect of MF was greatest in the topsoil (Fig. 9d). These findings indicate that soil infiltration patterns significantly alter the vertical profile partition of infiltrated water.

4. Discussion

4.1. Effect of grass root architecture on soil infiltration patterns

Root architecture serves as a critical biological driver of soil structure, which is a key determinant of hydrological functions such as water infiltration (Hao et al., 2020; Liu et al., 2020; Wang et al., 2023). This study demonstrates that grass cover significantly enhances soil infiltration capacity compared to clean tillage (CT) (Fig. 2). Notably, taproot species (GCt) produced a markedly greater improvement than

fibrous-root species (GCf). This enhancement was characterized not only by an increase in total infiltration volume but, more fundamentally, by a shift in the infiltration pattern. Under CT and GCf, matrix flow (MF) accounted for the majority (>60 %) of the cumulative infiltration amount. In contrast, preferential flow (PF) became the dominant flux under GCt, contributing 63.5 % of the total (Table 2). While the high velocity of PF at the pore scale is well-established, its contribution to the total volumetric flux at the plot scale depends on the effective cross-sectional area of connected pathways (Beven and Germann, 2013; Gao et al., 2018). Therefore, the observed dominance of MF under CT and GCf is both logical and consistent with documented phenomena in soils where PF pathways are sparse or discontinuous (Cullum, 2009; Jarvis et al., 2012; Xiang et al., 2019; Liao et al., 2025b). This study demonstrates that root architecture governs soil hydrological processes by determining the partitioning of water between PF and MF.

Different root architectures act as a "structural switch" controlling this partitioning by creating distinct soil conditions (Bodner et al., 2014; Blanco-Canqui and Ruis, 2020). A key feature in this orchard system (CT) is the compacted subsoil layer formed under long-term clean tillage management (Becerra et al., 2010; Ramos et al., 2022). This intact layer severely limited the vertical continuity of PF pathways (mean PF depth: 18.4 cm, Fig. 3), diverting infiltrating water into lateral matrix flow and resulting in absolute MF flux dominance (70.4 %) (Table 2). Although

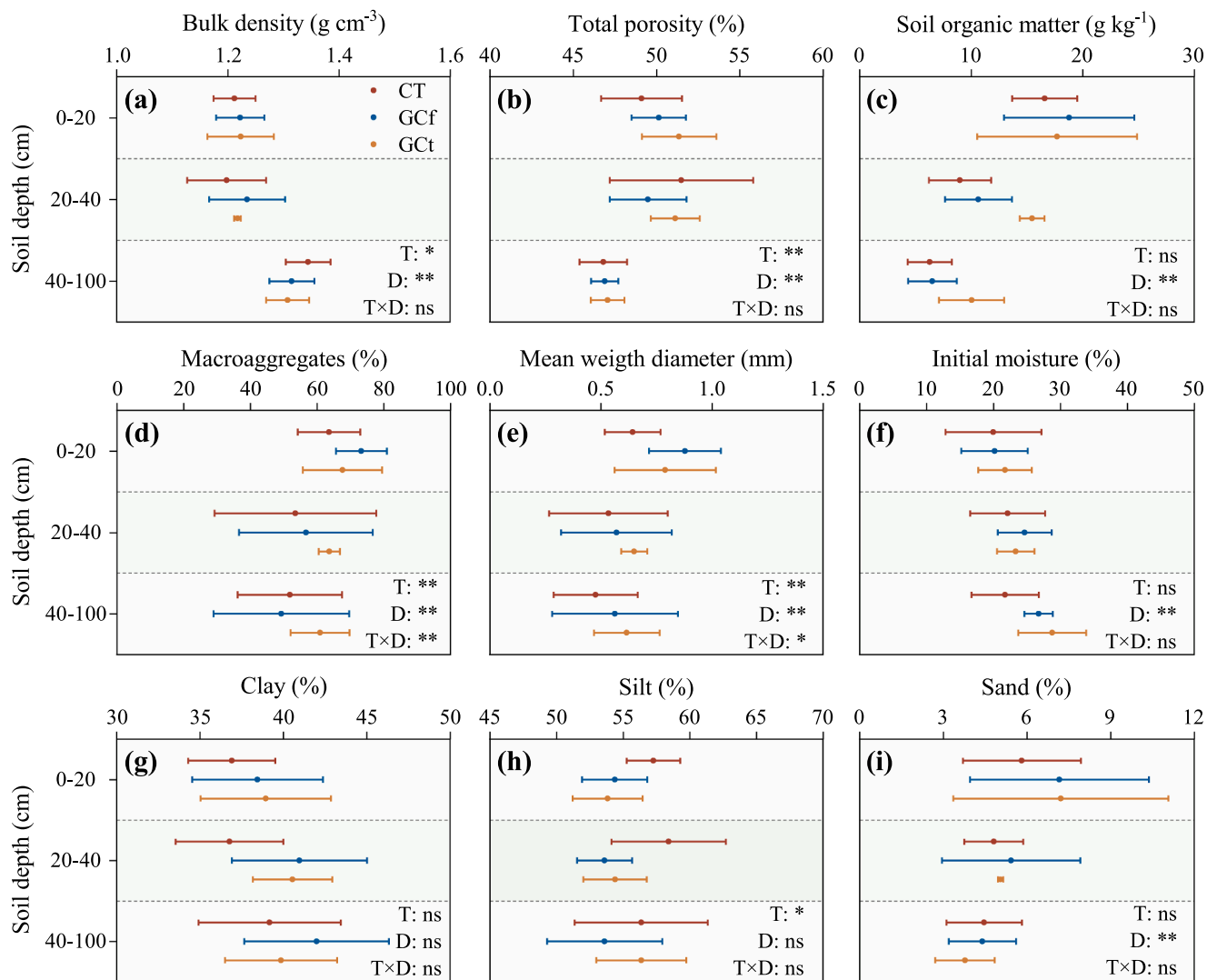


Fig. 7. Vertical variation of soil physicochemical properties under different grass cover treatments. Data are presented as means $\pm$ standard deviation (SD). T: treatment; D: depth; T $\times$ D: the interaction between treatment and depth. *, ** denote significance at $p < 0.05$ and $p < 0.01$, respectively; ns: not significant ($p > 0.05$). CT: clean tillage; GCf: fibrous-rooted grass cover; GCt: tap-rooted grass cover.

fibrous-root species (GCf) improved topsoil structure and increased initial PF flux, its roots failed to penetrate the compacted subsoil (Materechera et al., 1992; Han et al., 2015). Consequently, flow during mid-to-late infiltration was still impeded, leading to sustained MF flux dominance (60.9 %). This mechanism aligns with the findings in other soils featuring restrictive layers (Liao et al., 2025b; Du et al., 2025). In contrast, the taproot system of GCt breached the subsoil layer and established a stable network of vertical biopores (Zhang et al., 2022; Odone et al., 2024). This structural modification provided PF pathways with not only high velocity but also persistent and efficient hydraulic conductivity (Fig. 4c). Thus, beyond confirming the general benefit of grass cover on infiltration, this study quantitatively reveals the underlying mechanism—the shift from MF-dominant to PF-dominant flux partitioning—underlying this improvement.

This study provides a detailed analysis of the temporal dynamics between PF and MF, an aspect often overlooked (Hirashima et al., 2019). The timing of the transition between dominant flow regimes differed significantly among treatments (Fig. 4). In CT, degraded soil structure led to rapid saturation of limited PF paths, causing a very brief PF-dominant phase (5 min) before a swift transition to slow, MF-dominant infiltration (Yarahmadi et al., 2025). Fibrous-root species

(GCf) enhanced topsoil aggregation and organic matter, extended the PF-dominated phase to 16 min. Conversely, the taproot system of GCt maintained a connected macropore network that sustained PF throughout infiltration. These dynamics highlight that the PF/MF relationship is complex and time-dependent, serving as a sensitive indicator of soil hydraulic behavior. Crucially, taproot species not only increased PF flux but also extended its functional duration by stabilizing pore networks, thereby maximizing the window for efficient deep-water transport during rainfall (Guo et al., 2019; Cui et al., 2022; Odone et al., 2024). This insight reframes “improved infiltration” not as a static capacity but as a dynamic, temporally resilient process, with direct implications for capturing intense, long-duration rainfall events.

4.2. Impact of different infiltration patterns on soil water recharge

The partitioning between PF and MF during infiltration critically determines the subsequent redistribution and ultimate storage depth of soil water (Xiang et al., 2019; Lu et al., 2025). This study confirms that grass cover enhances soil water recharge (SWR) and, more importantly, elucidates how the two infiltration patterns differentially control water distribution within the soil profile. PF was identified as the key

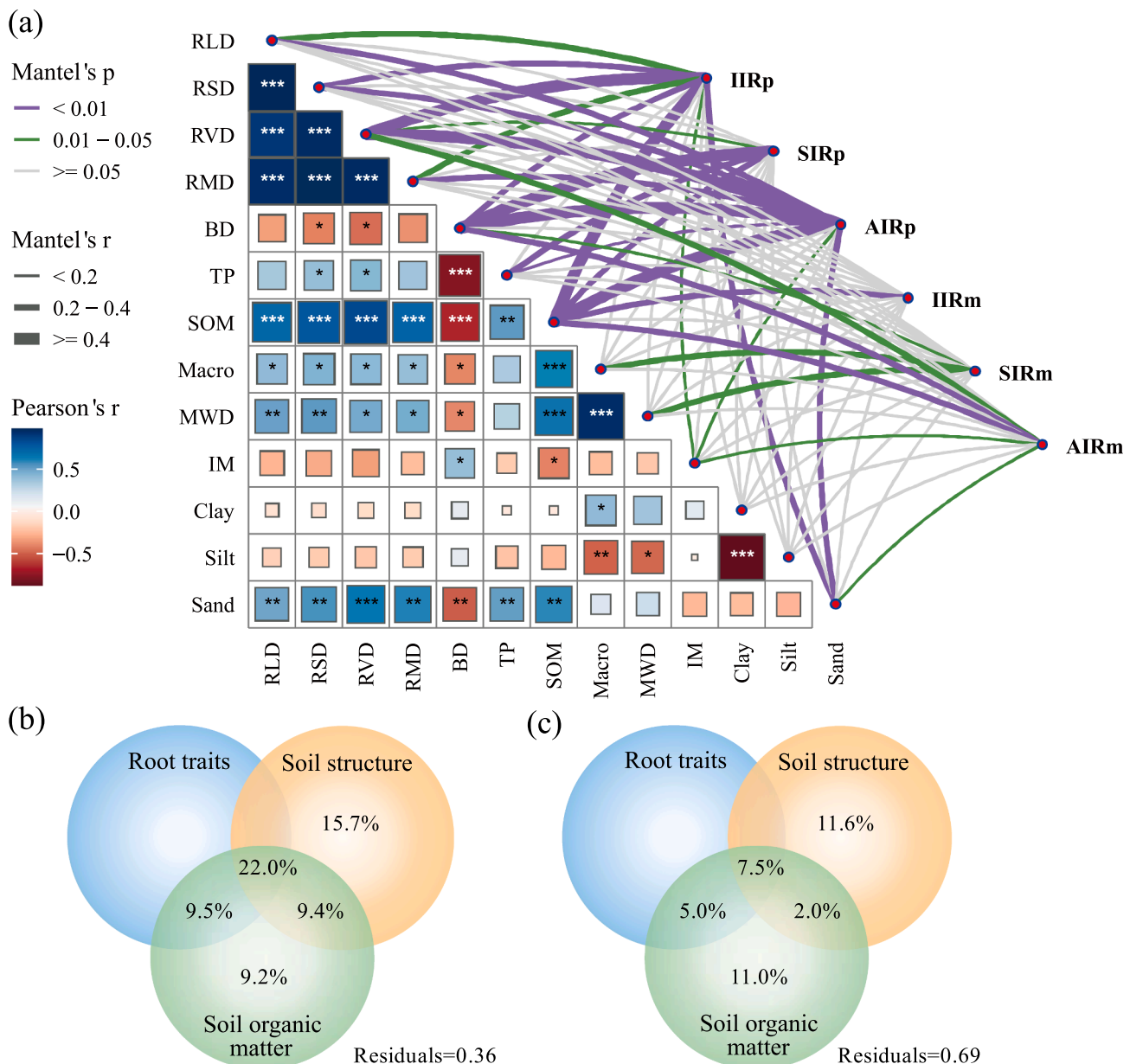


Fig. 8. Relationships between PF/MF infiltration rates and environmental factors based on Mantel-tests (a). Variance partitioning analysis (VPA) showing the relative contribution of root and soil properties to the variance in PF (b) and MF (c) infiltration rates. IIRp, SIRp, and AIRp represent the initial, steady, and average rate of PF infiltration; IIRm, SIRm, and AIRm represent the initial, steady, and average rate of MF infiltration. RLD: root length density; RSD: root surface area density; RVD: root volume density; RMD: root mass density; BD: bulk density; TP: total porosity; SOM: soil organic matter; Macro: macroaggregate; MWD: mean weight diameter; IM: initial moisture. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

mechanism for efficient water recharge of deep soil layers (Fig. 9b~c). Under the PF-dominant regime of GcT, water rapidly bypassed the shallow matrix via connected biopores (Cui et al., 2022; Lu et al., 2025), resulting in deep-layer SWR (40–100 cm) that was 4.00 times greater than under CT. The proportion of deep-layer recharge (PSWR) under GcT was also 1.42 and 1.54 times higher than under CT and GcF, respectively (Fig. 5b). In contrast, the MF-dominant regimes under CT and GcF promoted slow, capillarity-driven movement (Zhang et al., 2017; Zhao et al., 2022), concentrating recharge benefits in the topsoil and subsoil (0–40 cm) (Fig. 5b). Path analysis supported this mechanism, showing that root-mediated improvements in topsoil structure primarily enhanced MF and topsoil recharge, whereas deep soil recharge was predominantly governed by PF, which was strongly controlled by root traits (Fig. 9).

The PF/MF flux ratio ultimately governs the spatial fate of soil water,

leading to distinct ecohydrological functions (Fig. 10) (Tron et al., 2015; Mair et al., 2023). The MF-dominated regime acts as "sponge", storing water in root-dense topsoil where it is readily available for plant uptake but also susceptible to evaporation and rapid turnover (Zhang et al., 2017). Conversely, the PF-dominant regime functions as a "conduit", efficiently transferring water to depth while minimizing losses during transport, thereby creating a stable "soil water bank" (Cui et al., 2022; Lu et al., 2025). This functional divergence—'conduit' versus 'sponge'—directly links hydrological pathways to specific ecosystem services (Ajayi et al., 2021). It implies that management for "water conservation" requires balancing trade-offs: taproot systems (e.g., legumes) that promote PF enhance drought resilience and runoff reduction but may increase nutrient leaching risk, whereas fibrous-root systems (e.g., grasses) that promote MF optimize topsoil water and nutrient retention but may offer less drought protection. This framework shifts the

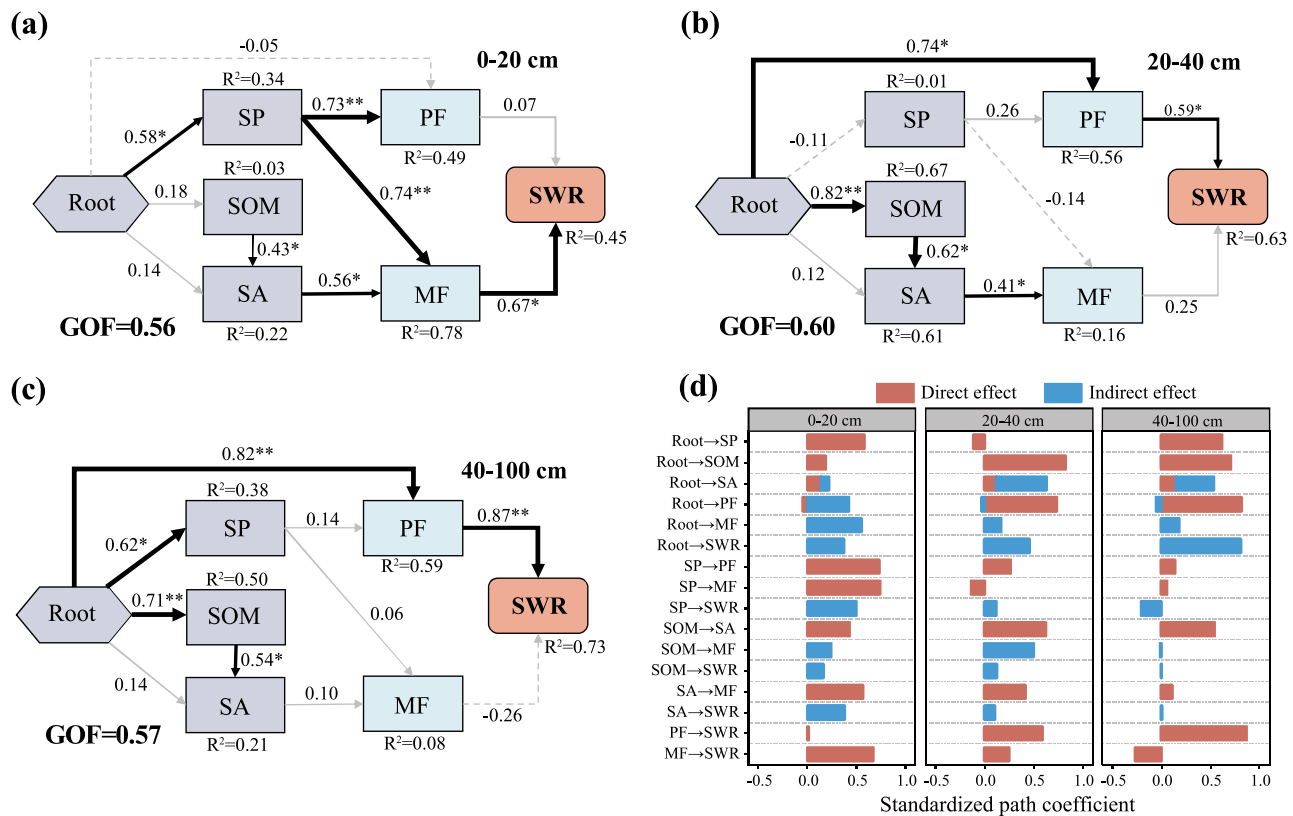


Fig. 9. Partial least squares path model illustrating the effects of the root characteristics on soil water recharge (SWR) in the topsoil (a), subsoil (b), and deep soil (c) layers. A partial least squares path model was conducted to examine the standardized path coefficients for the direct and indirect effects of variables (d). Numbers on arrows are standardized path coefficients, with solid and dashed lines indicating positive and negative correlations, respectively. The darkness or lightness of the lines denotes significance; and the thickness represents the magnitude of the effects. GOF: goodness-of-fit; SP: soil porosity; SOM: soil organic matter; SA: soil aggregates; PF, preferential flow; MF, matrix flow. $^*p < 0.05$; $^{**}p < 0.01$.

perspective from a simple water balance to a mechanistic understanding of flow pathways, providing a theoretical basis for targeted hydrological management through vegetation selection.

4.3. Limitations and implications

Although this study reveals the mechanisms linking root architecture to infiltration patterns and water redistribution, some limitations should be acknowledged. First, soil hydrological processes are scale-dependent. The PF-MF interactions quantified here at the plot scale may become more complex at hillslope or catchment scales due to increased subsurface heterogeneity (Beven and Germann, 2013). Although our relationships provide valuable modules for larger-scale models, direct extrapolation requires caution, and future multi-scale investigations are needed. Second, our findings are based on simulated infiltration under specific initial conditions. Infiltration capacity varies dynamically with seasons, wetting-drying cycles, and plant growth (Gao et al., 2018; Lu et al., 2020). A critical emerging question is how frequent drought-rewetting events under climate change might alter the PF/MF balance by affecting root dynamics and soil structure. Finally, while our method effectively quantifies the net flux of each domain, it does not resolve the small-scale, dynamic exchanges of water between PF and MF pathways in natural soils (Guo et al., 2025), which may affect predictions under extreme or transient conditions.

Despite these limitations, the findings offer significant theoretical and practical implications. Theoretically, this study establishes a coherent framework tracing cascading effects of root architecture on infiltration patterns and on water recharge. The quantified dynamics of PF provide critical parameters for improving the simulation of non-

equilibrium water flow in ecohydrological models (Gao et al., 2018; Mair et al., 2022). Practically, this framework provides a scientific basis for precision orchard groundcover management (Tron et al., 2015; Mair et al., 2023). To prioritize erosion control and surface water retention, fibrous-root species (e.g., gramineous crops) should be chosen to enhance MF-dominated "sponge" function. To enhance drought resilience and deep-water security, taproot species (e.g., leguminous crops) are preferable to promote the PF-dominated "conduit" function. For multifunctional landscapes, a mixed-species approach can optimally combine the 'conduit' and 'sponge' functions. This strategy leverages complementary hydrological processes to maximize orchard groundcover benefits in the face of water scarcity and climate variability.

5. Conclusion

This study elucidates the pivotal role of grass root architecture in regulating soil infiltration patterns and the redistribution of infiltrated water in orchard systems. The results demonstrate that grass cover significantly enhanced soil infiltration and water recharge capacity, with root architecture exerting a dual regulatory mechanism over soil hydrological pathways. Specifically, taproot grasses primarily reinforced and sustained preferential flow (PF), which facilitated deeper water recharge (40–100 cm). The deep recharge volume of taproot grasses was 4.00 times greater than that of clean tillage, accounting for 34.3 % of the total water recharge. This underscores the potential of taproot species to mitigate drought by establishing deep soil water reserves. In contrast, fibrous-root species exhibited matrix flow (MF) as the dominant process despite a prolonged PF-dominated phase. This flow pattern favored topsoil water retention (0–20 cm), achieving a topsoil

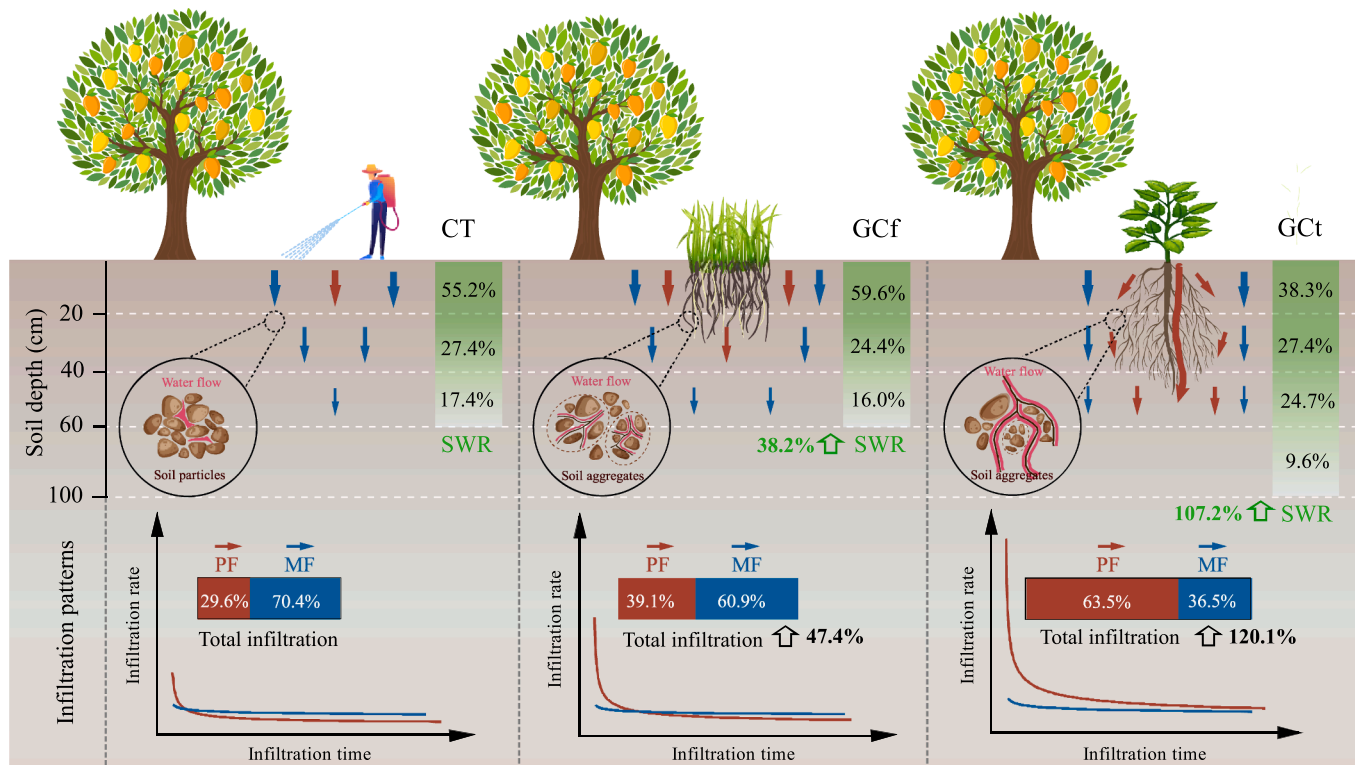


Fig. 10. Conceptual model of soil water recharge (SWR) resulting from grass root architecture-driven infiltration patterns. Green bars represent the proportional distribution of SWR across the soil profile. Percentages indicate the proportion of soil water recharge (PSWR) within each layer. Hollow green and black arrows denote the percentage increase in SWR and total infiltration, respectively, compared to clean tillage (CT). Solid red and blue arrows illustrate preferential flow (PF) and the matrix flow (MF) pathways. Gcf: fibrous-rooted grass cover; Gct: tap-rooted grass cover.

recharge volume 1.48 times that of clean tillage and representing 59.6 % of the total. These findings advocate a trait-based selection paradigm for groundcover management, where root architecture is strategically matched to target specific hydrological services. Taproot species can be utilized to enhance deep-water storage and drought resilience, whereas fibrous-root species contribute to topsoil moisture retention. Furthermore, mixed-species planting could synergize these functional benefits, offering a sustainable strategy for precision water management in orchard systems under climate uncertainty.

CRediT authorship contribution statement

Jie Yang: Visualization, Methodology, Funding acquisition, Data curation, Conceptualization. **Lingyun Wang:** Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Jian Duan:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yaojun Liu:** Visualization, Investigation, Formal analysis, Data curation, Conceptualization. **Chongjun Tang:** Writing – original draft, Methodology, Data curation, Conceptualization. **Haijin Zheng:** Methodology, Funding acquisition, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.still.2025.107045](https://doi.org/10.1016/j.still.2025.107045).

Data availability

Data will be made available on request.

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